STAT201 · STATISTICS · LECTURE 13

Regression Inference, Causation and Multiple Regression

Is the relationship found significant, how accurate are the estimates and predictions, and when regression does not answer the question “why”

Syllabus topic 13 · week 13 · 4 academic hours

LECTURE GOALS

- test the hypothesis $H_0 : \beta_1 = 0$ (t- and F-test) and construct a CI for the slope;
- construct a confidence interval for the mean response and a prediction interval for a new observation;
- interpret regression software output;
- distinguish correlation from causation; explain the role of omitted variables, reverse causation and Simpson’s paradox;
- understand the meaning of “all else equal” in multiple regression, R^2_{adj} and multicollinearity.

1 From the sample to the population

The coefficients b_0, b_1 (Lecture 12) are computed from a sample and fluctuate from sample to sample. As for a mean, we need to know their standard errors. When the conditions of the model hold (linearity, independence, constant variance, approximately normal errors):

$$\mathrm{SE}(b_1) = \frac{s_e}{\sqrt{S_{xx}}}, \qquad \mathrm{SE}(b_0) = s_e\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}, \qquad \frac{b_1 - \beta_1}{\mathrm{SE}(b_1)} \sim t_{n-2}. \tag{1.1}$$

The standard error of the slope decreases when n grows, when s_e decreases and when the values of x are more spread out (S_{xx} larger).

1.1 Testing the significance of the slope

$H_0 : \beta_1 = 0$ (no linear relationship) against $H_1 : \beta_1 \neq 0$:

$$t = \frac{b_1}{\mathrm{SE}(b_1)}, \qquad \nu = n - 2; \qquad \text{CI: } b_1 \pm t_{\alpha/2,\,n-2}\,\mathrm{SE}(b_1). \tag{1.2}$$

EXAMPLE 1. ADVERTISING AND SALES

From Lecture 12: $b_1 = 3.213$, $s_e = 2.195$, $S_{xx} = 45.33$. $\mathrm{SE}(b_1) = 2.195/\sqrt{45.33} = 0.326$. $t = 3.213/0.326 = 9.86$, $\nu = 4$, $p = 0.0006$. We reject H_0: the relationship is significant. 95% CI: $3.213 \pm 2.776 \cdot 0.326 = (2.31;\ 4.12)$ — a 1 million increase in advertising is associated with a sales increase of 2.3–4.1 million. For the intercept: $\mathrm{SE}(b_0) = 2.053$, $t = 3.96$, $p = 0.017$, CI $(2.42;\ 13.83)$.
Testing $H_0 : \beta_1 = 2$ (what if the return on advertising is 2): $t = (3.213 - 2)/0.326 = 3.72$, $p = 0.020$ — we reject: the return is significantly greater than 2.

1.2 The regression ANOVA table and the F-test

Source	SS	ν	MS	F
Regression	$SSR = 468.06$	1	468.06	**97.15**
Residual	$SSE = 19.27$	$n - 2 = 4$	$4.82 = s_e^2$	
Total	$SST = 487.33$	5		

$F = MSR/MSE$ tests the same hypothesis $\beta_1 = 0$; in simple regression $F = t^2$ ($9.86^2 = 97.15$) and the p-values coincide. The F-test becomes a test in its own right in multiple regression (significance of the model as a whole).

2 Intervals for prediction

At $x = x_0$ two questions are distinguished. (1) What is the *average* value of y for all objects with x_0? — the confidence interval for the mean response. (2) What will y be for a *new* object with x_0? — the prediction interval (it is wider because the random error ε is added):

$$\hat{y}_0 \pm t_{\alpha/2,n-2}\, s_e \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \quad \text{(mean)}, \tag{2.1}$$

$$\hat{y}_0 \pm t_{\alpha/2,n-2}\, s_e \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \quad \text{(individual)}. \tag{2.2}$$

EXAMPLE 2. PREDICTION AT $x_0 = 7$

$\hat{y}_0 = 30.62$. $\sqrt{1/6 + (7 - 5.667)^2/45.33} = 0.4538$, $s_e \cdot 0.4538 = 0.996$. The interval for the mean: $30.62 \pm 2.776 \cdot 0.996 = (27.85;\ 33.38)$. The prediction interval: $\sqrt{1 + 0.2059} = 1.098$, $2.195 \cdot 1.098 = 2.410$; $30.62 \pm 2.776 \cdot 2.410 = (23.93;\ 37.31)$. The interval for the mean is narrowest at $\bar{x}$ and grows toward the edges: at $x_0 = 10$ the half-width for the mean is ± 4.64 against ± 2.48 at $x_0 = \bar{x} = 5.67$.

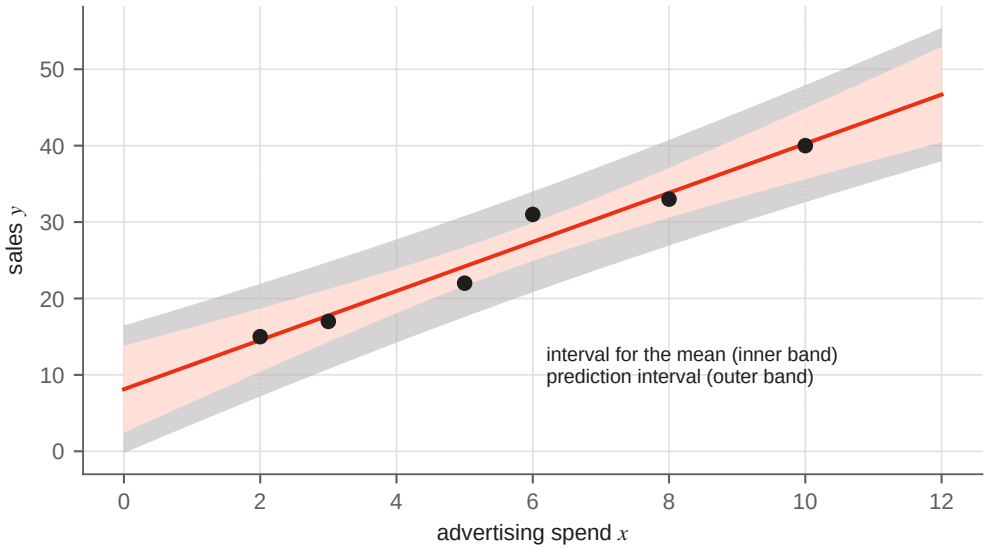


Fig. 1. The regression line, the 95% CI for the mean response (inner band, narrowing near $\bar{x}$) and the 95% prediction interval for a new observation (outer band).

COMMON MISTAKE

A prediction interval is not reducible to a CI for the mean: even with an infinite sample, individual values remain scattered around the line by $\pm 1.96\,\sigma$. Do not confuse "the average for a group" with "the value for a particular customer".

3 Reading software output

Typical output (Excel, R, Python) contains: coefficients, their standard errors, t-statistics and p-values, R^2, s_e, F and the p-value of F. For our example: Intercept 8.125 (SE 2.053, $t = 3.96$, $p = 0.017$); Slope 3.213 (SE 0.326, $t = 9.86$, $p = 0.0006$); $R^2 = 0.960$; $s_e = 2.195$; $F = 97.15$ (Significance $F = 0.0006$), $n = 6$. The ability to "read" such output in a minute is a standard exam requirement.

4 Correlation and causation

The significance of the slope says only that the linear relationship in the data is unlikely to be a coincidence. It does not mean that a change in x causes a change in y. Reasons for false conclusions:

- **An omitted (lurking) variable** Z affects both quantities. Example: the more firefighters at a fire, the greater the damage — but both quantities are determined by the scale of the fire. Ice-cream sales and the number of drownings rise together because of hot weather.
- **Reverse causation**: banks with higher profits spend more on advertising — and not only does advertising raise profit.
- **Coincidence** (especially when searching through many variables).
- **Simpson's paradox**: a relationship in the pooled data reverses sign within groups.

EXAMPLE 3. SIMPSON'S PARADOX

A call centre tests two conversation scripts with customers (success = the customer took out a product).

	Simple customers	Difficult customers	Total
Script A	81/87 (93.1%)	192/263 (73.0%)	273/350 (78.0%)
Script B	234/270 (86.7%)	55/80 (68.8%)	289/350 (82.6%)

In each group script A is better, but in total B looks better: script A was used more often with difficult customers (263 of 350), and B with simple ones (270 of 350). The customer type is a confounding variable: it is related both to the choice of script and to the result. The correct conclusion is drawn taking the groups into account.

REMEMBER

Causal conclusions are reliable when the values of x are assigned *randomly* (a randomized experiment, an A/B test): then, by construction, x is unrelated to omitted factors. With observational data one needs (a) control of confounding variables, (b) a justification of the direction of causation and (c) cautious wording: "is associated with", not "leads to".

5 Multiple regression (an introduction)

A model with several explanatory variables: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$. The coefficient β_j shows the change in the mean of y when x_j grows by one unit *with the other variables held constant* (ceteris paribus).

EXAMPLE 4. ADVERTISING, PRICE AND SALES

10 weeks: x_1 — advertising spend, x_2 — price, y — sales.

Week	1	2	3	4	5	6	7	8	9	10
x_1	2	3	5	6	8	10	4	7	9	1
x_2	12	11	11	10	9	9	12	10	9	13
y	20	24	31	38	47	52	26	41	48	15

Regression on x_1 alone: $\hat{y} = 11.0 + 4.22x_1$, $R^2 = 0.989$. Multiple regression: $\hat{y} = 40.86 + 3.19\,x_1 - 2.29\,x_2$, $R^2 = 0.996$, $R^2_{\text{adj}} = 0.995$, $s_e = 0.944$. Interpretation: at a constant price an additional million of advertising gives on average $+3.19$ in sales (not 4.22: the simple regression "attributed" to advertising part of the price effect, because in weeks with more advertising the price was lowered, $r(x_1, x_2) = -0.95$); at constant advertising a price increase of one unit reduces sales by 2.29. Both coefficients are significant ($p = 0.00003$ and $p = 0.014$). The prediction for $x_1 = 6$, $x_2 = 10$: 37.2.

REMEMBER

R^2 does not decrease when variables are added, so $R^2_{\text{adj}} = 1 - \dfrac{SSE/(n-k-1)}{SST/(n-1)}$ is used to compare models. A strong correlation among the explanatory variables (*multicollinearity*) makes the estimates unstable: large SE and "jumping" coefficients. Categorical factors are entered through indicators (0/1): the coefficient on an indicator is the difference in means between the category and the base one, all else equal.

6 Key terms

Term	*In Russian*	Meaning
Standard error of the slope	*Стандартная ошибка наклона*	$s_e/\sqrt{S_{xx}}$
Slope t-test	*t-тест наклона*	$t = b_1/\operatorname{SE}(b_1)$, $n-2$ df
Prediction interval	*Интервал предсказания*	for a new observation
CI for the mean response	*Доверительный интервал отклика*	for the mean of y at x_0
Confounder	*Искажающая переменная*	related to both x and y
Simpson's paradox	*Парадокс Симпсона*	the relationship reverses sign within groups
Multiple regression	*Множественная регрессия*	several predictors
Multicollinearity	*Мультиколлинеарность*	strong relationship among predictors

Formulas of Lecture 13

$$\operatorname{SE}(b_1) = \frac{s_e}{\sqrt{S_{xx}}}, \quad t = \frac{b_1}{\operatorname{SE}(b_1)}\ (\nu = n-2), \quad b_1 \pm t_{\alpha/2}\operatorname{SE}(b_1); \quad F = \frac{SSR/1}{SSE/(n-2)} = t^2;$$

$$\hat{y}_0 \pm t_{\alpha/2} s_e \sqrt{\tfrac{1}{n} + \tfrac{(x_0-\bar{x})^2}{S_{xx}}}\ \text{(mean)}; \qquad \hat{y}_0 \pm t_{\alpha/2} s_e \sqrt{1 + \tfrac{1}{n} + \tfrac{(x_0-\bar{x})^2}{S_{xx}}}\ \text{(individual)}.$$

7 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] In a regression on $n = 27$ observations $b_1 = 2.4$, $\operatorname{SE}(b_1) = 0.6$. Test $H_0 : \beta_1 = 0$ ($\alpha = 0.05$) and construct a 95% CI for β_1.

Problem 2 [A] The sample correlation is $r = 0.45$ with $n = 30$. Test $H_0 : \rho = 0$ using $t = r\sqrt{n-2}/\sqrt{1-r^2}$ ($\alpha = 0.05$).

Problem 3 [A] For each statement name a probable lurking variable or an alternative explanation: (a) cities with more churches have more crime; (b) employees who have had more training earn more; (c) firms with high advertising have larger profits.

Problem 4 [B] Study-time data: $x = 1, 2, 3, 4, 5$, $y = 50, 55, 65, 70, 85$ (Lecture 12): $b_1 = 8.5$, $s_e = 3.03$, $S_{xx} = 10$.

(a) Find $\mathrm{SE}(b_1)$ and test the significance of the slope.
(b) Construct a 95% CI for β_1.
(c) Test $H_0 : \beta_1 = 5$ against $H_1 : \beta_1 \neq 5$.

Problem 5 [B] For the data of the previous problem, at $x_0 = 4.5$ hours find the 95% CI for the mean score and the 95% prediction interval for a new student. Explain the difference.

Problem 6 [B] Apartment data (Lecture 12): $\hat{y} = 5.79 + 0.776x$, $s_e = 1.576$, $n = 5$, $S_{xx} = 2320$, $\bar{x} = 66$.

(a) Test the significance of the slope and construct a 95% CI.
(b) Find the 95% CI for the mean price of 70 m^2 apartments and the prediction interval for the price of a particular 70 m^2 apartment.

Problem 7 [B] In the output of a regression on $n = 27$ observations it is known that: $SSR = 160$, $SSE = 90$, $b_1 = 2.4$.

(a) Fill in the ANOVA table (ν, MS, F).
(b) Find R^2 and s_e.
(c) Find the t-statistic of the slope and $\mathrm{SE}(b_1)$.

Problem 8 [B] Loan applications are submitted at two branches. Approved: branch X — small loans $18/20$, large loans $24/80$; branch Y — small $68/80$, large $5/20$. Find the approval rates by group and overall. Which branch is "better"? Explain the paradox.

Problem 9 [B] An online shop introduced free delivery for some customers, choosing those who order more often. A month later their average bill rose by 12% relative to the rest. Can we say that free delivery increased the bill? How should a study be organized to draw a causal conclusion?

Problem 10 [B] A model of weekly sales: $\hat{y} = 20 + 3x_1 - 1.5x_2$ (x_1 — advertising, x_2 — price); $n = 20$, $R^2 = 0.80$.

(a) Interpret the coefficients.
(b) Find the prediction at $x_1 = 8$, $x_2 = 10$.
(c) Find R^2_{adj}.

Problem 11 [C] Derive $\mathrm{Var}(b_1) = \sigma^2/S_{xx}$ using the representation $b_1 = \sum c_i y_i$, where $c_i = (x_i - \bar{x})/S_{xx}$, and the y_i are independent with variance σ^2.

Problem 12 [C] Show that $\mathrm{Var}(\hat{y}_0) = \sigma^2 \left(\dfrac{1}{n} + \dfrac{(x_0 - \bar{x})^2}{S_{xx}} \right)$ and explain where the "1+" in the prediction interval comes from.